

trabalho2

December 12, 2024

```
[51]: %matplotlib inline

import numpy as np
from matplotlib import pyplot as plt
from qiskit import QuantumCircuit
from qiskit.primitives import StatevectorSampler
from qiskit.quantum_info import Operator
from qiskit.quantum_info import Statevector

sampler = StatevectorSampler()
```

0.0.1 IIQC Trabalho 2, Miguel Curto Castela, uc2022212972

0.0.2 2.1 Exercícios simples 1

1. Escrever o produto Kronecker (produto tensorial) dos qubits:

a) $|0\rangle \otimes |1\rangle$

b) $|0\rangle \otimes |+\rangle$

c) $|+\rangle \otimes |1\rangle$

d) $|-\rangle \otimes |+\rangle$

tendo em conta que:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Resposta ao exercício anexada na última página do notebook em PDF e no PDF exercícios_manuscritos_2022212972.pdf incluído na entrega

2. Escreva o estado: $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle$ como dois qubits separados.

Resposta ao exercício anexada na última página do notebook em PDF e no PDF exercícios_manuscritos_2022212972.pdf incluído na entrega

0.0.3 2.2 Exercícios simples 2

1. Calcule a unitária (U) de um circuito de um qubit criado pela sequência de portas: $U = XZH$. Utilize o Qiskit para verificar os seus resultados.

Resposta ao exercício anexada na última página do notebook em PDF e no PDF `exercicios_manuscritos_2022212972.pdf` incluído na entrega

[52]: *#verificação feita com Qiskit*

```
qc = QuantumCircuit(1)
qc.x(0)
qc.z(0)
qc.h(0)

unitary = Operator(qc).data
print("unitary matrix:")
print(unitary)
```

unitary matrix:

```
[[-0.70710678+0.j  0.70710678+0.j]
 [ 0.70710678+0.j  0.70710678+0.j]]
```

2. Considere um circuito diferente com as mesmas portas, mas “empilhadas” para ter um circuito com 3 qubits, aplicando cada porta a um qubit diferente. Calcule o seu produto kronecker (produto tensorial) para obter a unitária (U) do circuito de 3 qubits, em seguida, verifique a sua resposta utilizando o Qiskit.

Resposta ao exercício anexada na última página do notebook em PDF e no PDF `exercicios_manuscritos_2022212972.pdf` incluído na entrega

[53]: *#verificação feita com Qiskit*

```
qc = QuantumCircuit(3)
qc.x(2)
qc.z(1)
qc.h(0)

unitary = Operator(qc).data
print("unitary matrix:")
print(unitary)
```

unitary matrix:

```
[ [ 0.          +0.j  0.          +0.j  0.          +0.j  0.          +0.j
   0.70710678+0.j  0.70710678+0.j  0.          +0.j  0.          +0.j]
 [ 0.          +0.j  0.          +0.j  0.          +0.j  0.          +0.j
   0.70710678+0.j -0.70710678+0.j  0.          +0.j  0.          +0.j]
 [ 0.          +0.j  0.          +0.j  0.          +0.j  0.          +0.j
   0.          +0.j  0.          +0.j -0.70710678+0.j -0.70710678+0.j]
 [ 0.          +0.j  0.          +0.j  0.          +0.j  0.          +0.j
   0.          +0.j  0.          +0.j -0.70710678+0.j  0.70710678+0.j]
 [ 0.70710678+0.j  0.70710678+0.j  0.          +0.j  0.          +0.j
   0.          +0.j  0.          +0.j  0.          +0.j  0.          +0.j]]
```

```
[ 0.70710678+0.j -0.70710678+0.j  0.          +0.j  0.          +0.j
  0.          +0.j  0.          +0.j  0.          +0.j  0.          +0.j]
[ 0.          +0.j  0.          +0.j -0.70710678+0.j -0.70710678+0.j
  0.          +0.j  0.          +0.j  0.          +0.j  0.          +0.j]
[ 0.          +0.j  0.          +0.j -0.70710678+0.j  0.70710678+0.j
  0.          +0.j  0.          +0.j  0.          +0.j  0.          +0.j]]
```

0.0.4 Exercícios simples 3

1. Crie um circuito quântico que produza o estado de Bell: $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$. Utilize o Qiskit para construir o circuito e statevector para verificar o seu resultado.

```
[54]: #Bell state: |psi = 1/sqrt(2) (|01 + |10 )
#Circuit to create the Bell state
expectedBellState = Statevector([0, 1, 1, 0]) / np.sqrt(2)

qc = QuantumCircuit(2)
qc.x(1)
qc.h(0)
qc.cx(0, 1)

print(qc)

statevector = Statevector.from_instruction(qc)
print("circuit statevector", statevector)

if statevector.equiv(expectedBellState):
    print("O circuito produz o bell state esperado")
else:
    print("O circuito não produz o bell state esperado")
```

q_0: H

q_1: X X

```
circuit statevector Statevector([0.          +0.j, 0.70710678+0.j, 0.70710678+0.j,
  0.          +0.j],
  dims=(2, 2))
```

O circuito produz o bell state esperado

2. O circuito que criou acima transforma o estado $|00\rangle$ em $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$, calcule a matriz unitária deste circuito utilizando o Qiskit. Verifique se esta unitária realiza de facto a transformação correta.

```
[55]: #Matriz unitária do circuito
unitary = Operator(qc).data

print("unitary matrix:")
```

```

print(unitary)

initial_state = Statevector.from_label('00')
print("initial state:", initial_state)

final_state = initial_state.evolve(unitary)
print("final state:", final_state)

if statevector.equiv(expectedBellState):
    print("A matriz unitária produz o bell state esperado")
else:
    print("A matriz unitária não produz o bell state esperado")

```

unitary matrix:

```

[[ 0.          +0.j  0.          +0.j  0.70710678+0.j  0.70710678+0.j]
 [ 0.70710678+0.j -0.70710678+0.j  0.          +0.j  0.          +0.j]
 [ 0.70710678+0.j  0.70710678+0.j  0.          +0.j  0.          +0.j]
 [ 0.          +0.j  0.          +0.j  0.70710678+0.j -0.70710678+0.j]]

```

```

initial state: Statevector([1.+0.j, 0.+0.j, 0.+0.j, 0.+0.j],
                           dims=(2, 2))

```

```

final state: Statevector([0.          +0.j, 0.70710678+0.j, 0.70710678+0.j,
                          0.          +0.j],
                        dims=(2, 2))

```

A matriz unitária produz o bell state esperado

0.0.5 2.4 Algoritmo Deutsch-Jozsa

Apresente a sua implementação, simulação de alguns casos, e execução em hardware real de pelo menos dois casos aproveitando o número de qubits disponíveis.

```

[56]: import numpy as np
import matplotlib.pyplot as plt
from qiskit.visualization import plot_histogram
from qiskit import QuantumCircuit, transpile
from qiskit_aer import Aer
from qiskit.visualization import plot_histogram
from qiskit_aer import AerSimulator
from qiskit_ibm_runtime import QiskitRuntimeService
import os
from dotenv import load_dotenv

if "IBM_TOKEN" in os.environ:
    del os.environ["IBM_TOKEN"]
load_dotenv()
IBM_TOKEN = os.getenv("IBM_TOKEN")
#print(f"IBM_TOKEN: {IBM_TOKEN}")
service = QiskitRuntimeService(channel="ibm_quantum", token=IBM_TOKEN)
backend = service.least_busy(operational=True, simulator=False)

```

```
sim = AerSimulator()
print(backend)
```

```
<IBMQBackend('ibm_kyiv')>
```

```
[57]: def constantOracle(len):
        oracle = QuantumCircuit(len+1)

        oracle.barrier()

        output = np.random.randint(2)
        if(output == 1):
            oracle.x(len)

        oracle.barrier()

        return oracle

def balancedOracle(len):
    oracle = QuantumCircuit(len+1)

    oracle.barrier()

    #Random implementation
    b = np.random.randint(1, 2**len)
    b_str = format(b, '0'+str(len)+'b')

    for qb in range(len):
        if(b_str[qb] == '1'):
            oracle.x(qb)

    for qb in range(len):
        oracle.cx(qb, len)

    for qb in range(len):
        if(b_str[qb] == '1'):
            oracle.x(qb)

    oracle.barrier()

    #Static implementation
    '''
    oracle = QuantumCircuit(len+1)

    oracle.barrier()
```

```

    for qb in range(len):
        if(qb % 2 == 0):
            oracle.x(qb)

    oracle.barrier()

    for qb in range(len):
        oracle.cx(qb, len)

    oracle.barrier()

    for qb in range(len):
        if(qb % 2 == 0):
            oracle.x(qb)

    oracle.barrier()
'''

return oracle

```

```

[58]: def createOracle(len):
    constant = constantOracle(len)
    balanced = balancedOracle(len)

    return constant, balanced

```

```

[59]: def createCircuit_dj(len, oracle):
    dj = QuantumCircuit(len+1, len)

    #input qubits
    for qb in range(len):
        dj.h(qb)

    #output qubit
    dj.x(len)
    dj.h(len)

    #apply the oracle
    dj = dj.compose(oracle)

    #apply Hadamard gates to the first n qubits
    for qb in range(len):
        dj.h(qb)

    #measure the first n qubits
    for qb in range(len):
        dj.measure(qb, qb)

```

```

    return dj

def simulateCircuit_dj(dj):
    job = sim.run(dj)
    results = job.result()
    graph = results.get_counts()
    plot_histogram(graph)

def runCircuit_dj(dj):
    transpiledCircuit = transpile(dj, backend, optimization_level=3)
    transpiledCircuit.draw('mpl', idle_wires=False)

    job = backend.run([transpiledCircuit], shots=1024)
    job.job_id()

    results = job.result()
    graph = results.get_counts()

    plot_histogram(graph)

def jobResults(job_id):
    job = service.job(job_id)
    results = job.result()
    graph = results.get_counts()
    plot_histogram(graph)
    #plt.show()

#número de qubits (no Deutsch-Jozsa e no grover)
len = 4

#create the oracles
constantOracle, balancedOracle = createOracle(len)

```

```

[60]: #Deutsch-Jozsa circuit with balanced oracle
balancedDj = createCircuit_dj(len, balancedOracle)

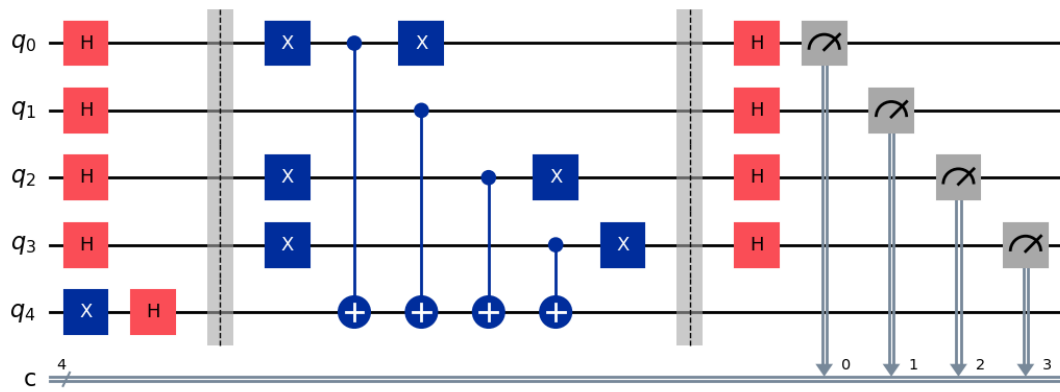
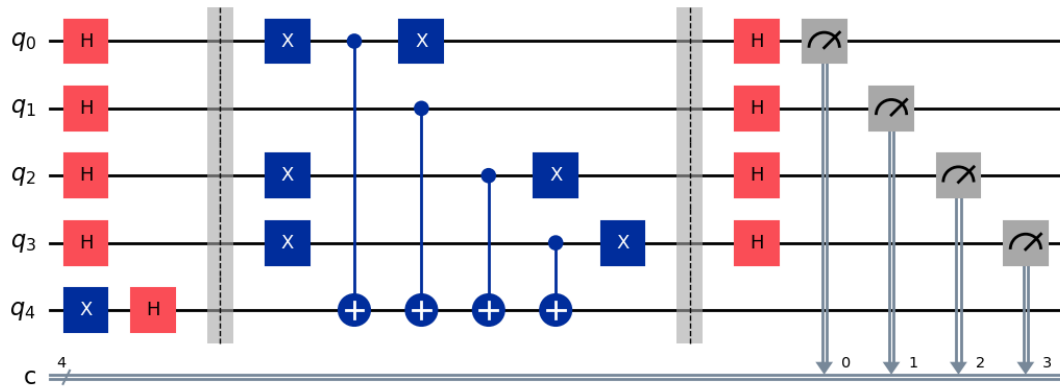
#run it on real hardware
#runCircuit_dj(balancedDj)

print("Deutsch-Jozsa circuit with balanced oracle ")
balancedDj.draw('mpl', idle_wires=False)

```

Deutsch-Jozsa circuit with balanced oracle

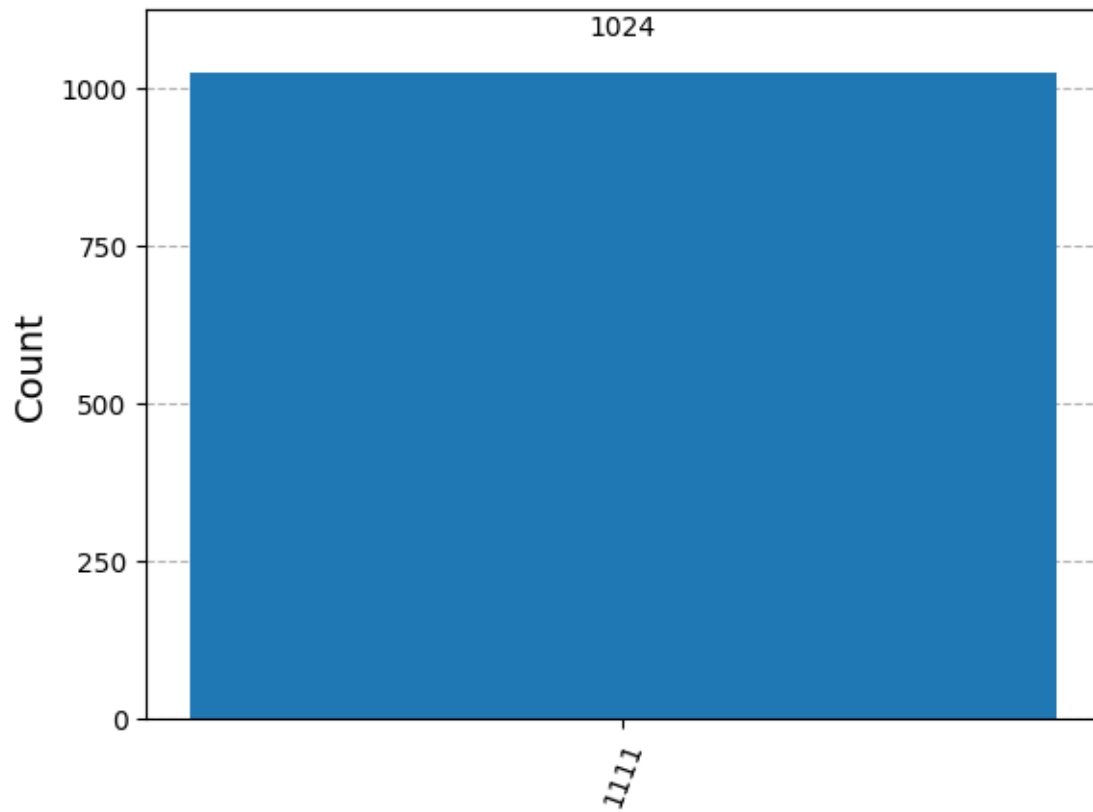
[60]:



```
[61]: #print("simulation of the Deutch-Jozsa circuit with balanced oracle ")
      #print(balancedDj)
```

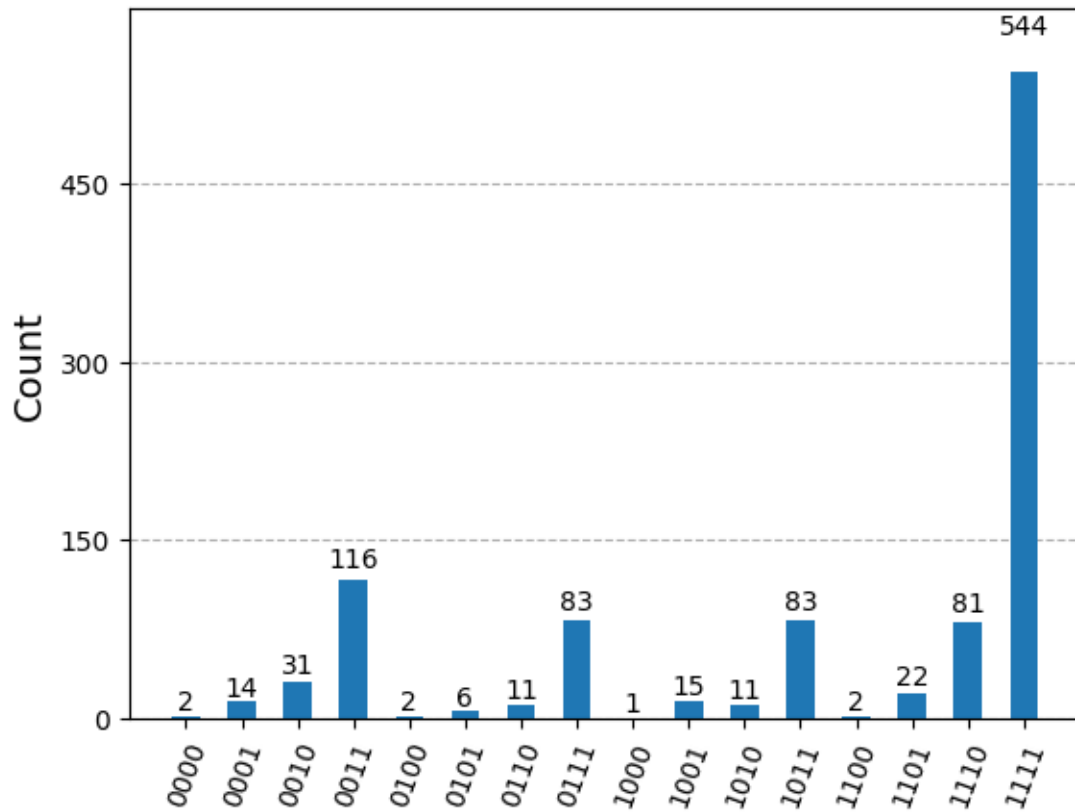
```
[62]: print("results of the simulation")
      simulateCircuit_dj(balancedDj)
```

results of the simulation



```
[63]: print("balanced iracle job results:")  
      jobResults("cxde3j6vw7kg008atm10")
```

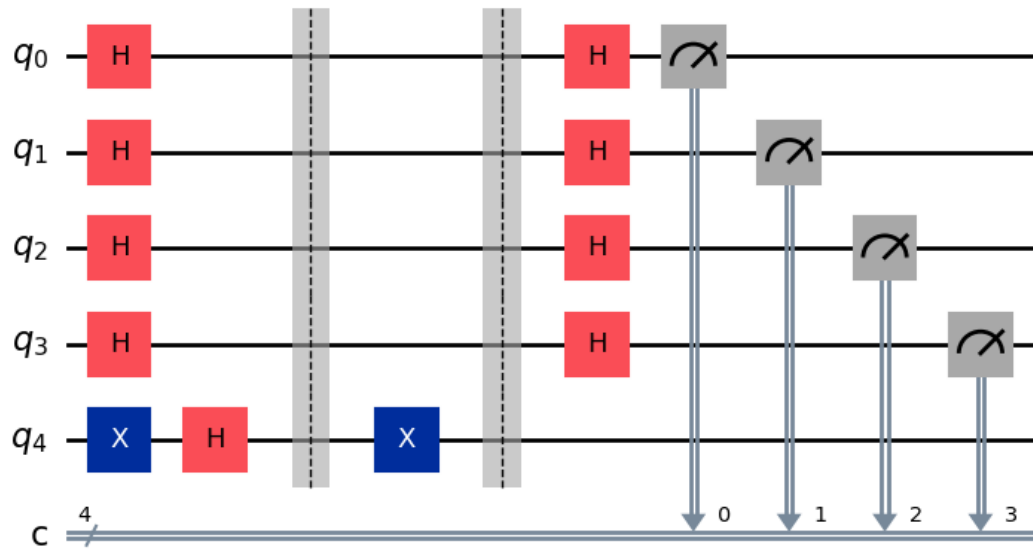
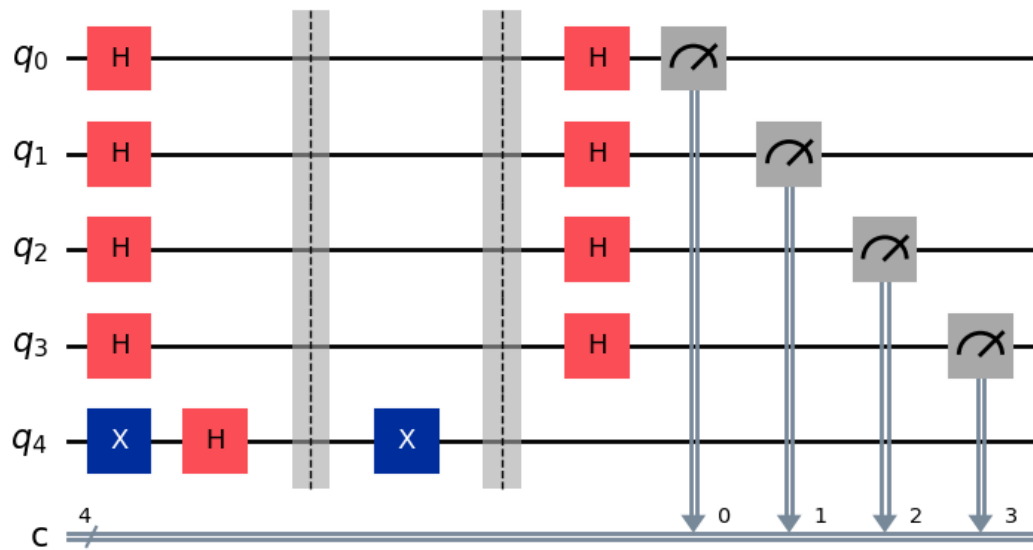
balanced iracle job results:



```
[64]: #Deutsch-Jozsa circuit with the constant oracle
      constantDj = createCircuit_dj(len, constantOracle)
      # run it on real hardware
      #runCircuit_dj(constantDj)
      print("Deutsch-Jozsa circuit with constant oracle ")
      constantDj.draw('mpl', idle_wires=False)
```

Deutsch-Jozsa circuit with constant oracle

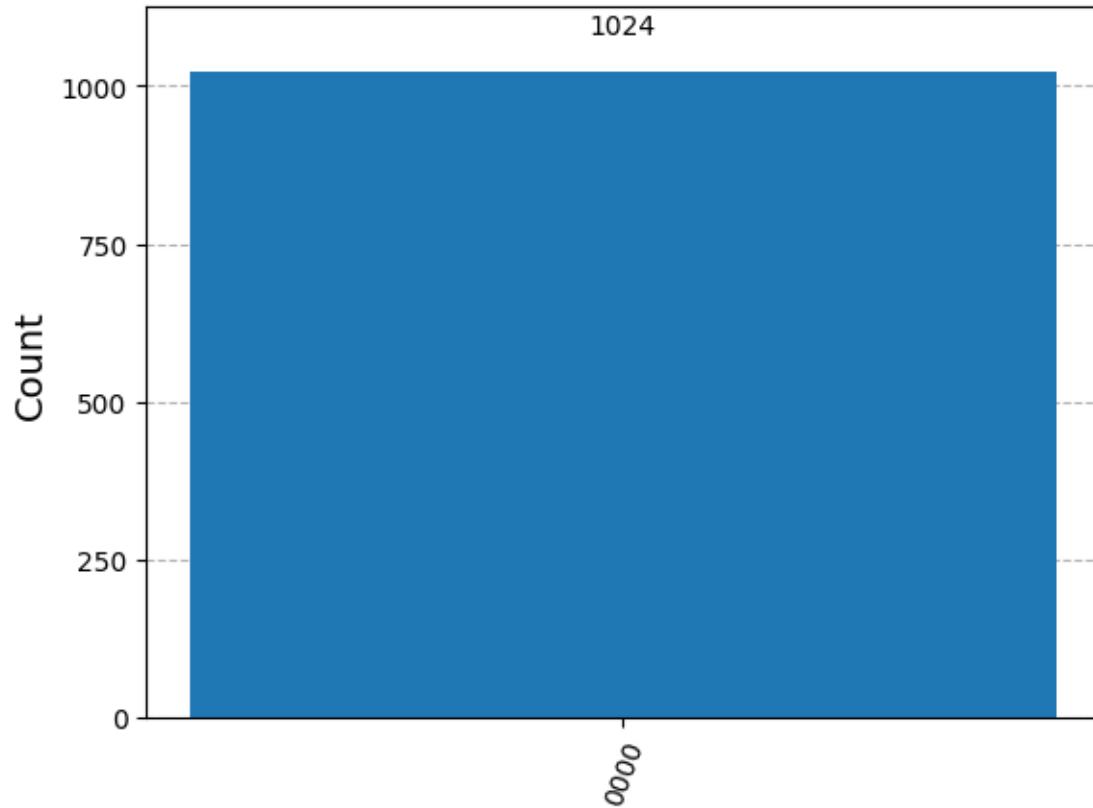
[64]:



```
[65]: #print("simulation of the Deutsch-Jozsa circuit with constant oracle ")
      #print(constantDj)
```

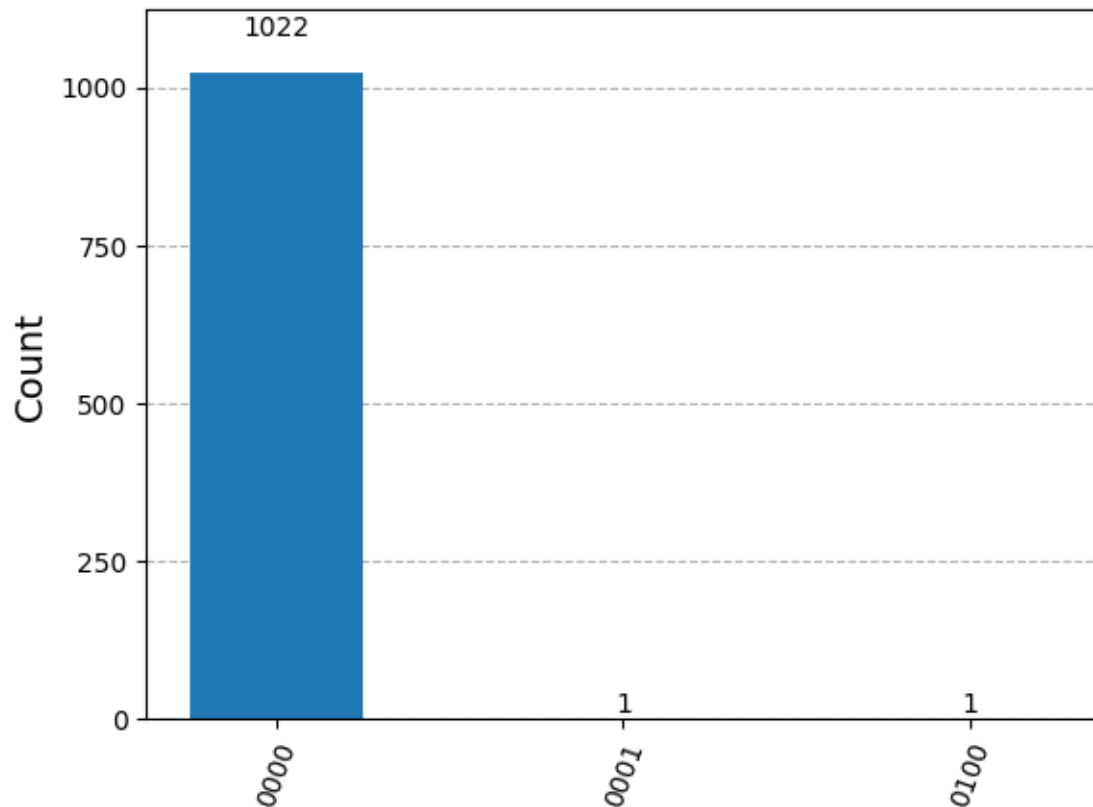
```
[66]: print("results of the simulation")
      simulateCircuit_dj(constantDj)
```

results of the simulation



```
[67]: print("constant oracle job results:")  
      jobResults("cxdejd3ej4g0089x87g")
```

constant oracle job results:



0.0.6 2.5 Grover's Algorithm

Apresente a sua implementação, simulação de alguns casos, e execução em hardware real de pelo menos dois casos aproveitando o número de qubits disponíveis.

```
[68]: def createPhaseOracle_1():
    #states: |0101>, |0110>, |1100>, |1111>
    oracle1 = QuantumCircuit(len)
    oracle1.barrier()
    oracle1.cz(0, 2)
    oracle1.cz(1, 2)
    oracle1.cz(2, 3)
    oracle1.barrier()
    return oracle1

def createPhaseOracle_2():
    #states: |0110>, |0111>, |1100>, |1101>
    oracle2 = QuantumCircuit(len)
    oracle2.barrier()
    oracle2.cz(1, 2)
    oracle2.cz(2, 3)
```

```
oracle2.barrier()
return oracle2
```

```
[69]: def createDiffuser():
    diffuser = QuantumCircuit(len)
    diffuser.barrier()

    for qubit in range(len):
        diffuser.h(qubit)

    for qubit in range(len):
        diffuser.x(qubit)

    diffuser.h(len-1)
    diffuser.mcx(list(range(len-1)), len-1)
    diffuser.h(len-1)

    for qubit in range(len):
        diffuser.x(qubit)

    for qubit in range(len):
        diffuser.h(qubit)
    diffuser.barrier()

    return diffuser
```

```
[70]: def createCircuit1_grover():
    grover1 = QuantumCircuit(len)

    for qubit in range(len):
        grover1.h(qubit)

    #target states
    grover1 = grover1.compose(createPhaseOracle_1(), list(range(len)))

    grover1 = grover1.compose(createDiffuser(), list(range(len)))
    grover1.measure_all()

    return grover1

def createCircuit2_grover():
    grover2 = QuantumCircuit(len)

    for qubit in range(len):
        grover2.h(qubit)

    #target states
```

```

grover2 = grover2.compose(createPhaseOracle_2(), list(range(len)))

grover2 = grover2.compose(createDiffuser(), list(range(len)))
grover2.measure_all()

return grover2

```

```

[71]: def runCircuit_grover(grover):
    transpiledGrover = transpile(grover, backend)
    transpiledGrover.draw('mpl', idle_wires=False)

    job = backend.run([transpiledGrover], shots=1024)
    job.job_id()

    results = job.result()
    graph = results.get_counts()
    plot_histogram(graph)

```

```

[72]: def simulateCircuit_grover(grover):
    job = sim.run(grover)
    result = job.result()
    graph = result.get_counts()
    plot_histogram(graph)

```

```

[73]: #create grover circuit (with oracle 1)
grover_1 = createCircuit1_grover()

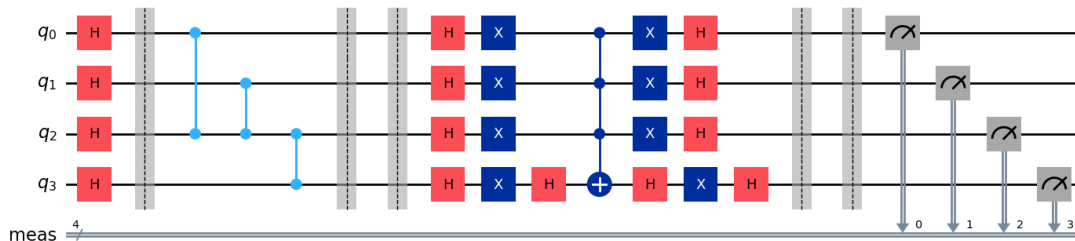
#run it on real hardware
#runCircuit_grover(grover_1)

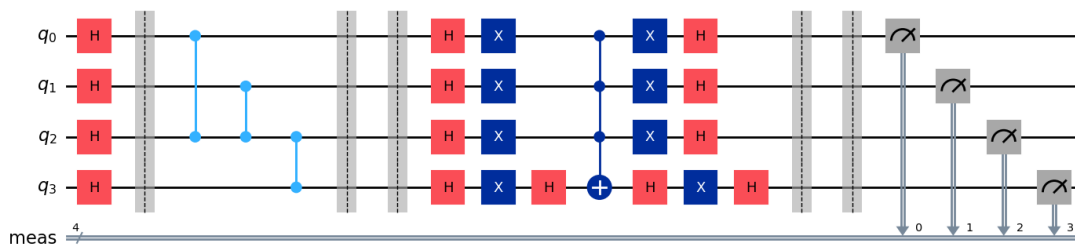
print("grover_1 circuit for the first oracle:")
grover_1.draw('mpl', idle_wires=False)

```

grover_1 circuit for the first oracle:

[73]:

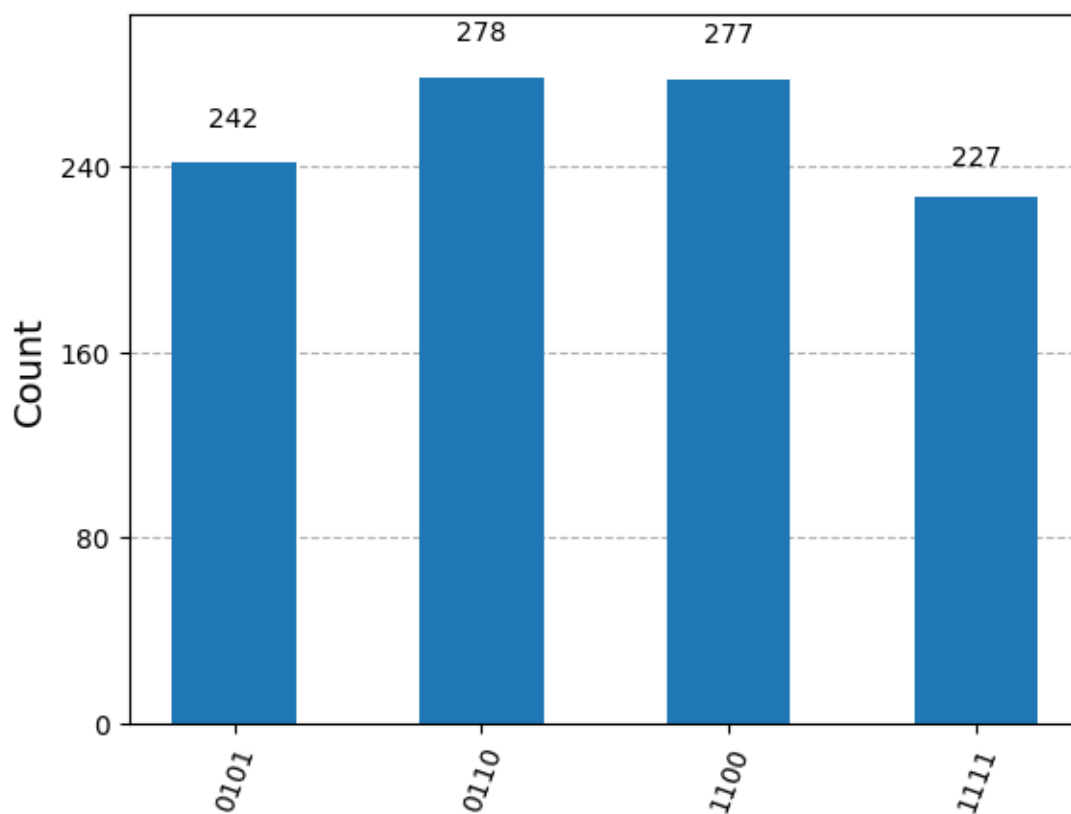




```
[74]: #print("grover_1 circuit simulation:")
      #print (grover_1)
```

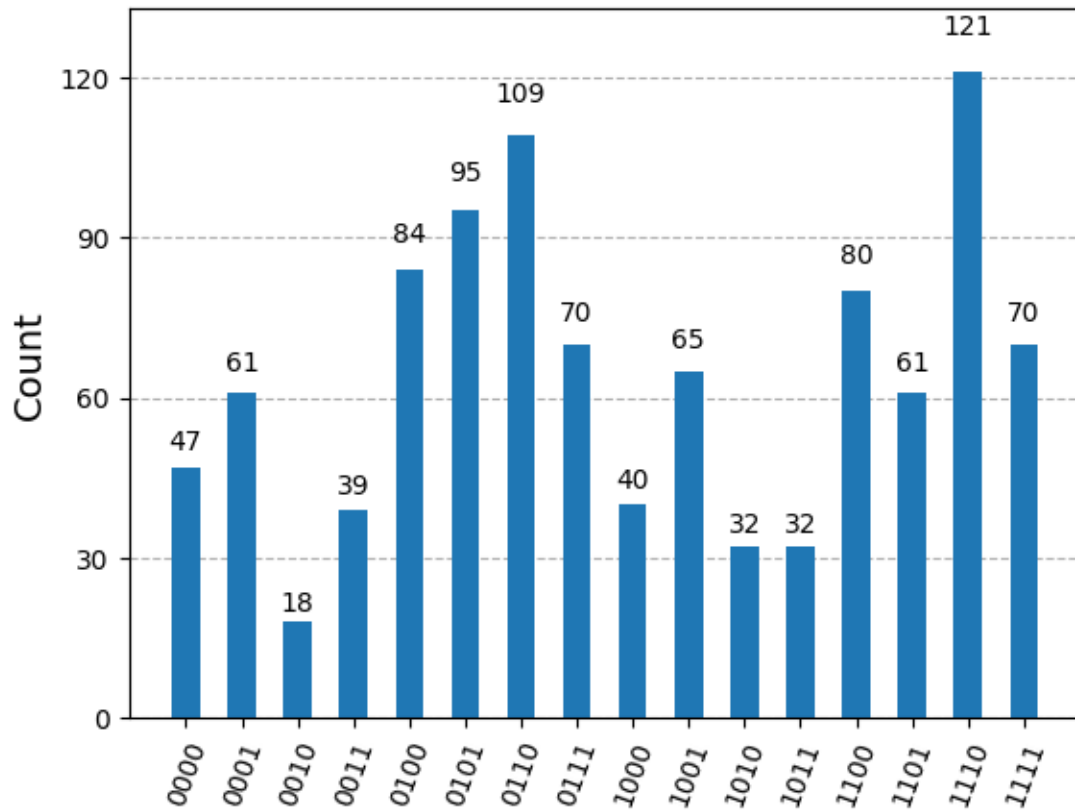
```
[75]: print("grover_1 simulation results:")
      simulateCircuit_grover(grover_1)
```

grover_1 simulation results:



```
[76]: print("grover_1 job results:")
      jobResults("cxdejvkgcckg008bgve0")
```


grover_1 job results:



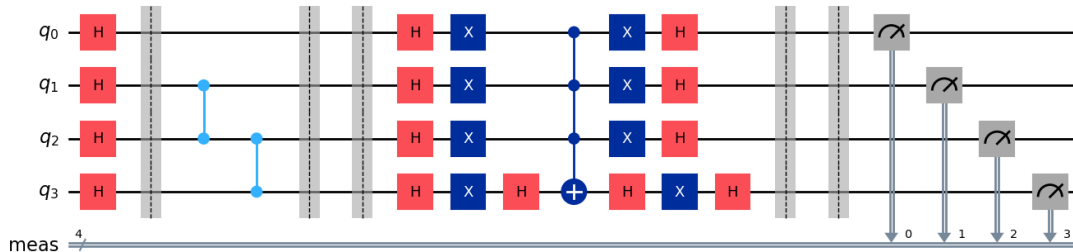
```
[77]: # create grover circuit (with oracle 2)
grover_2 = createCircuit2_grover()

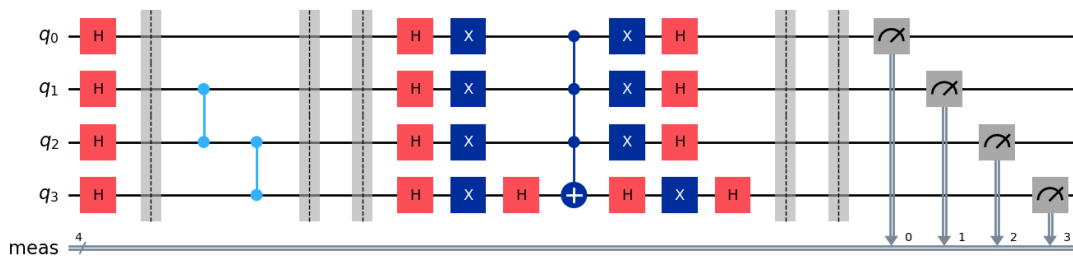
#run it on real hardware
#runCircuit_grover(grover_2)

print("grover_2 circuit for the second oracle:")
grover_2.draw('mpl', idle_wires=False)
```

grover_2 circuit for the second oracle:

[77]:

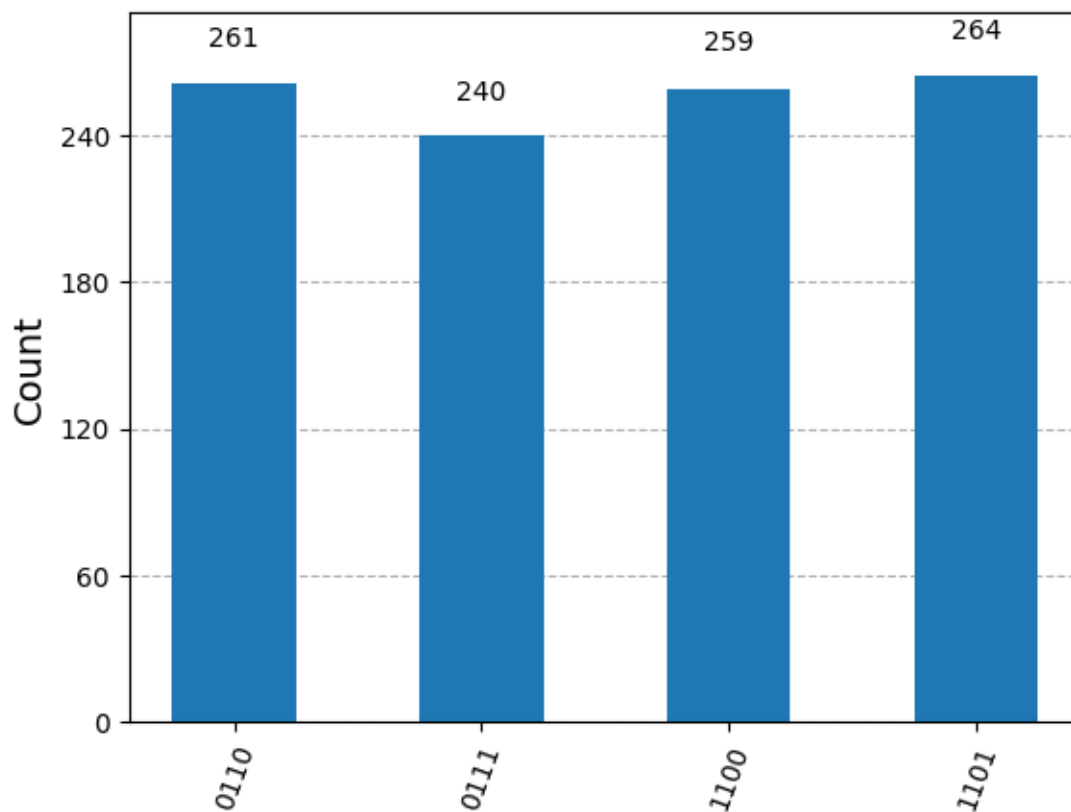




```
[78]: #print("grover_2 circuit simulation:")
      #print (grover_2)
```

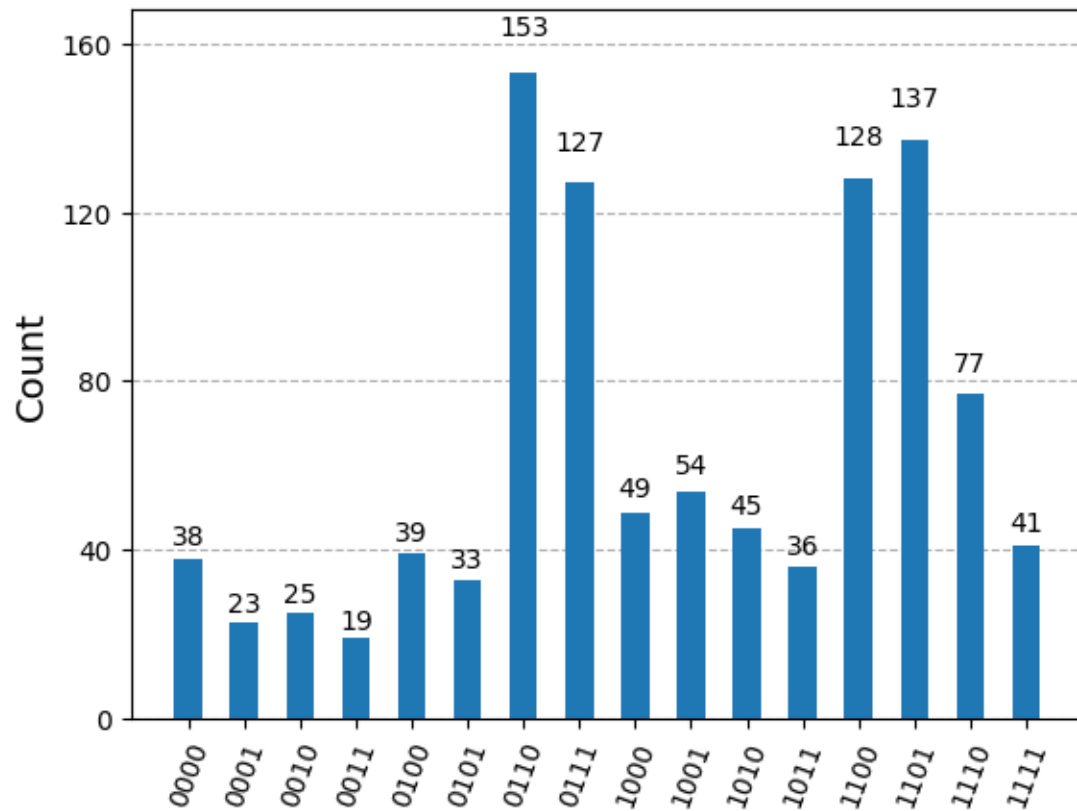
```
[79]: print("grover_2 simulation results:")
      simulateCircuit_grover(grover_2)
```

grover_2 simulation results:



```
[80]: print("grover_2 job results:")  
      jobResults("cxdek2cfdnwg008s0s00")
```

grover_2 job results:



11QL 2º bndado

$$2.1 \quad |+\rangle = 1/\sqrt{2} (|0\rangle + |1\rangle) = 1/\sqrt{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$|-\rangle = 1/\sqrt{2} (|0\rangle - |1\rangle) = 1/\sqrt{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$a) |0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{bmatrix} 1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{bmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$b) |0\rangle \otimes |+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1/\sqrt{2} \begin{bmatrix} 1 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ 0 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{bmatrix} = 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$c) |+\rangle \otimes |1\rangle = 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1/\sqrt{2} \begin{bmatrix} 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{bmatrix} = 1/\sqrt{2} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

$$d) |-\rangle \otimes |+\rangle = 1/\sqrt{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \otimes 1/\sqrt{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1/2 \begin{bmatrix} 1 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ -1 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{bmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

$$2) \quad |\psi\rangle = 1/\sqrt{2} |00\rangle + 1/\sqrt{2} |01\rangle \text{ como 2 qubits separados}$$

$$|\psi\rangle = |0\rangle \otimes (1/\sqrt{2} |0\rangle + i/\sqrt{2} |1\rangle)$$

2.2

1)

matriz (U) de um circuito de um qubit criada pelas portas $U = XZH$

$$H = 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$X \times Z \times H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \times 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = 1/\sqrt{2} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

2)

$$U = X \otimes Z \otimes H = \begin{bmatrix} 0 \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} & 1 \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \\ 1 \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} & 0 \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \otimes 1/\sqrt{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & -1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$